SE 216 – SOFTWARE PROJECT MANAGEMENT
PROJECT RISKS DOCUMENT

PROJECT NAME: NeuroMatter: Universal Local AI Hub

GROUP MEMBERS: Duru Deniz Alptekin, Şevval Acar, Kaan Efe Koygun, Ege Haktan Aykut, Ali Utku Aydın, Eren Karlı, Deren Kocabaş, Efe Berkay Soyucak

LIKELIHOOD RANK	RISK DESCRIPTION
1	**AI Processing Time -** High probability that local AI models (YOLOv8n, Qwen2-VL) will exceed the 2-second threshold; if processing exceeds this, the system fails to prevent resident sensory overload, its core mission.
2	**Device Compatibility -** Potential difficulties in ensuring seamless communication between different brands of Matter-compatible devices; limited compatibility restricts the "Universal" aspect of the hub.
3	**Software Synchronization -** Technical challenges in synchronizing Flutter UI with C++/Python control layers and SQLCipher; failure results in system downtime, leaving users without safety support.
4	**Storage Consumption -** Risk of sensor data from InfluxDB consuming local storage and RAM faster than anticipated; may cause slight delays in automation but does not compromise immediate safety-critical functions.
5	**AI Accuracy -** Probability of the AI failing to maintain 95% detection accuracy in complex home environments; low accuracy could lead to household accidents for neurodivergent residents.
6	**Power Failure -** Risk of hardware failure during extended power outages; total power loss renders all safety-critical monitoring and response functions inactive.
7	**Data Security -** Low probability of 256-bit AES encryption being compromised in a local-only environment; however, exposure of sensitive behavioral logs violates the primary promise of data sovereignty.
8	**System Overload -** Low probability of system failure when managing 20+ concurrent smart devices; performance drops in Grafana visualization without affecting the core AI logic or response.

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IMPACT RANK	RISK DESCRIPTION
1	**AI Processing Time -** High probability that local AI models (YOLOv8n, Qwen2-VL) will exceed the 2-second threshold; if processing exceeds this, the system fails to prevent resident sensory overload, its core mission.
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LIKELIHOOD RANK	IMPACT RANK	COMBINED RANK	RISK DESCRIPTION
1	1	2	**AI Processing Time -** High probability that local AI models (YOLOv8n, Qwen2-VL) will exceed the 2-second threshold; if processing exceeds this, the system fails to prevent resident sensory overload, its core mission.
3	4	7	**Software Synchronization -** Technical challenges in synchronizing Flutter UI with C++/Python control layers and SQLCipher; failure results in system downtime, leaving users without safety support.
2	6	8	**Device Compatibility -** Potential difficulties in ensuring seamless communication between different brands of Matter-compatible devices; limited compatibility restricts the "Universal" aspect of the hub.
5	3	8	**AI Accuracy -** Probability of the AI failing to maintain 95% detection accuracy in complex home environments; low accuracy could lead to household accidents for neurodivergent residents.
7	2	9	**Data Security -** Low probability of 256-bit AES encryption being compromised in a local-only environment; however, exposure of sensitive behavioral logs violates the primary promise of data sovereignty.
6	5	11	**Power Failure -** Risk of hardware failure during extended power outages; total power loss renders all safety-critical monitoring and response functions inactive.
4	7	11	**Storage Consumption -** Risk of sensor data from InfluxDB consuming local storage and RAM faster than anticipated; may cause slight delays in automation but does not compromise immediate safety-critical functions.

8	**8**	**16**	**System Overload -** Low probability of system failure when managing 20+ concurrent smart devices; performance drops in Grafana visualization without affecting the core AI logic or response.